

YAN KÉVIN ZÉ

Curriculum Vitae

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Nationality: Cameroonian

EDUCATION

MSc in Mathematical Sciences

2025 – 2026

African Institute for Mathematical Sciences (AIMS), Kigali, Rwanda

Thesis Title: On Stallings' Fibration Theorem for Knot Exterior.

Thesis Summary: Studied Stallings' fibration theorem for knot exteriors, relating the algebraic structure of the knot fundamental group to fibrations over the circle via Seifert surfaces and infinite cyclic coverings.

Relevant Courses: Graphs, Matroids, and Designs; Partial Differential Equations; Topological Data Analysis; Python Programming.

Master 2 in Mathematics (Geometry)

2023 – 2025

University of Yaoundé I, Cameroon

Thesis Title: Cohomology of Homogeneous Spaces: Stiefel and Complex Grassmann Manifolds.

Thesis Summary: Computation of singular cohomology of Stiefel manifolds and complex Grassmannians, using algebraic topology tools including the Gysin exact sequence, Euler and Chern classes.

Relevant Courses: Algebraic Topology; De Rham Cohomology; Algebraic Geometry I; Lie Groups and Lie Algebras; Symplectic Geometry I and II.

Master 1 in Pure Mathematics (Algebra and Geometry)

2022 – 2023

University of Yaoundé I, Cameroon

Bachelor's Degree in Fundamental Mathematics

2021 – 2022

University of Yaoundé I, Cameroon

Baccalauréat C

2018 – 2019

Cameroon

SKILLS

- Languages: French (fluent), English (working proficiency), Bikélé (mother tongue, fluent)
- Software: Python, R, Mathematica, LaTeX, Microsoft Word, Excel

EXPERIENCE

Research Intern

July – September 2026

MIORPA Research Programme, University of Oxford, United Kingdom

- Working on new supergravity solutions of the form $\text{AdS}_3 \times M(7)$ encoding the near-horizon region of a black hole
- Using toric geometry methods and differential equations in Mathematica, supervised by Tabea Sieper and Chris Couzens

Tutor

2023, 2024

University of Yaoundé I, Cameroon

- Taught Linear Algebra I and II
- Taught Ordinary Differential Equations

Tutor

2018, 2022, 2023, 2024

High School, Cameroon

- Taught Mathematics and Physics
- Levels: Terminale C and Première C

AWARDS

- 2023–2024: Award for Academic Excellence
- 2015–2016, 2017–2018: Best High School Student Award

REFERENCES

Prof. Ana García Lecuona, Thesis Supervisor, University of Glasgow, Ana.Lecuona@glasgow.ac.uk

Prof. Blaise Tchapnda, Academic Director, African Institute for Mathematical Sciences, Kigali, Rwanda,
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Dr. Hilaire Mbiakop Georges, Supervisor, University of Yaoundé I, Cameroon

Dr. Donald Woukeng Feudjio, Academic Tutor, African Institute for Mathematical Sciences, Kigali, Rwanda,
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Dr. Olakunle Abawonse, Post Doctoral Researcher, African Institute for Mathematical Sciences, Kigali, Rwanda,
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